Gravity as Relational Difference Elimination: A Harmonic Closure Derivation

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Model Status. This manuscript is an exploratory, falsifiable framework. All claims are labeled throughout using the taxonomy in Table 1. No experimental validation independent of the author's prior program is claimed. A relativistic completion does not yet exist.

Version 4.0 advances: (1) the n-selection argument is now structured as a generative hierarchy with a mathematical irreducibility condition; (2) Newton's constant is reexpressed as $G = \alpha \cdot R_V^2 \cdot a_m/m_p$, where $a_m = c^2/R_m$ is the centripetal acceleration at the confinement radius — giving G a clear three-factor physical interpretation; (3) α_G is reinterpreted not as a free parameter but as the confinement-frequency ratio $\alpha_G \approx (16/15)(\omega_m/\omega_{\mathrm{Pl}})^2$, placing it within the framework's own frequency hierarchy; (4) the Planck length emerges from a gravitational self-consistency condition $\eta = 1$ of the torsional field; (5) interpretive language is softened throughout to reflect one possible reading of the formalism.

Claim Type	Description
[POSTULATE]	A foundational assumption — not derived, not yet falsified.
[DERIVED]	A mathematical consequence of the postulates.
[CALIBRATION]	A parameter fixed by matching to an empirical value.
[HYPOTHESIS]	A testable physical conjecture requiring external validation.
[SPECULATIVE]	An exploratory program note. Not yet a genuine prediction.

Table 1: Claim-type legend used throughout the manuscript.

Abstract

We propose a relational-informational model in which gravity emerges as the macroscopic tendency of physical systems to eliminate differences between their projected relational states. The framework rests on a universal harmonic partition of system frequencies $\omega_N^2 = \omega_V^2 + \omega_m^2$, previously applied within the same research program to hadronic, atomic, and cosmological systems.

The distance between two systems is identified as a relational phase offset — the accumulated difference in identity-mode frequency, measured in units of the identity radius R_{id}. Gravity is the macroscopic expression of systems tending to reduce this offset. Conservation of relational information over isotropic surfaces yields the inverse-square law as a derived result, not an assumption.

The gravitational source is the relational phase-offset density $\rho_{\mathrm{rel}} = \kappa\omega_m^2\rho_m$, where $\kappa = R_V^2 R_m/m_p$ is the relational closure volume. This leads to $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$ with $G = \alpha\kappa\omega_m^2$.

A central structural result is that the framework defines a natural gravitational scale constructed entirely from m_p, c, $\hbar$, and the $n = 4$ closure, without gravitational input:

$$\frac{G}{\alpha} = \kappa\omega_m^2 = 4\sqrt{15}\,\frac{\hbar c}{m_p^2}.$$

The dimensionless coupling α is the relational ratio of the visible mode within the closure: $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N\omega_m)$. G and α are not independently free — both are linked through this scale, which is fixed by the harmonic partition alone. A physically transparent reexpression of G is:

$$G = \alpha \cdot \frac{R_V^2}{m_p} \cdot a_m, \qquad a_m = \frac{c^2}{R_m},$$

separating G into three factors: the relational transmission probability α, the specific identity cross-section R_V^2/m_p, and the intrinsic confinement acceleration a_m.

Version 4.0 also reinterprets α_G. Rather than reading $\alpha_G = Gm_p^2/(\hbar c)$ as a free empirical input, the framework supports reading it as the ratio of the confinement frequency to the Planck frequency: $\alpha_G \approx (16/15)(\omega_m/\omega_{\mathrm{Pl}})^2$. Under this reading, α_G measures the gravitational instability of the torsional field at the hadronic closure scale, and the Planck length emerges as the self-consistency threshold of that field. This does not yet constitute a derivation of α_G from first principles, but it eliminates its status as an arbitrary constant within the framework.

Version 4.0 advances the treatment of n-selection: the value $n = 4$ is now motivated as the first closure depth at which the confinement-to-visible ratio $2^n - 1$ is arithmetically irreducible — a product of two distinct primes — making the two modes genuinely incommensurable. This is embedded in a generative hierarchy ($n = 0$ through 4) that maps natural physical systems to successive closure depths.

Keywords: emergent gravity; relational physics; harmonic partition; binary closure; phase offset; inverse-square law; unification; weak equivalence principle.

Originality and Novelty Statement

What is novel. This work identifies a natural gravitational scale $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$ constructed entirely from hadronic constants and the $n = 4$ binary closure, without gravitational input. The dimensionless coupling α is expressed as the relational ratio of the visible mode: $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N \omega_m)$. Together these show that G and α are not independent free parameters but are linked through the harmonic partition.

Previous emergent-gravity approaches (Verlinde [4], Jacobson [3], Padmanabhan [5]) reproduce Newtonian gravity qualitatively from thermodynamic analogies but leave G as a proportionality constant absorbed into entropy definitions. This model expresses G in terms of microscopically meaningful quantities and identifies the structural origin of the gravitational hierarchy.

How it differs from existing literature. The key distinction from thermodynamic approaches is structural: entropic and holographic frameworks assign temperature and entropy to macroscopic screens, but have no microscopic mechanism that determines the numerical value of G. Verlinde derives an entropic force $F = T\Delta S/\Delta x$ that recovers Newton's second law near a holographic screen, but G remains absorbed into the Bekenstein–Hawking entropy formula as a prior. The present framework assigns a specific relational closure volume $\kappa = R_V^2 R_m/m_p$ to each nucleon's closure state — the product of an identity cross-section R_V^2 and a confinement depth R_m — making G explicitly dependent on hadronic-scale physics and therefore in principle sensitive to composition.

Acknowledged limitations.

- Weak-field, stationary regime only.
- The coupling α is calibrated, not derived (one free parameter).
- Relativistic tensor completion remains future work.
- Isotropy of Postulate 4 is stated as consistent with closure symmetry but not yet derived from the generative algebra.
- The n-selection argument is a structural motivation, not a proof.
- Prior applications of the harmonic partition are internal consistency results from the same research program and are not claimed as independent validations.

1 Introduction

Gravity remains the only fundamental interaction whose coupling constant G is imposed empirically rather than derived from deeper principles. While General Relativity provides an exquisite classical description, its conceptual origin — why mass curves spacetime with precisely this strength — remains unexplained. The question this paper addresses is: can the value of G, and the inverse-square form of the gravitational force, be expressed as consequences of pre-existing microscopic structure, rather than treated as primitive inputs?

Within the author's research program, a harmonic partition of system frequencies has been applied to three domains (Table 2). This paper extends the same principle to gravity, adding two structural results: the inverse-square law and Newton's constant, both emerging from the harmonic partition without separate assumptions.

Scale	**System**	**Harmonic Ratio**	**Result**
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1 : 15$	$r_p \approx 0.8412$ fm
Atomic	Valence stability (s–p)	$a : b$ prime or 2^n	Harmonic weight $P = a + b$
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ when $R = R_H$

Table 2: Prior applications of the relational projection framework. Internal consistency results, not independent validations.

2 Relational Foundations

2.1 The Harmonic Partition Postulate

[POSTULATE] 1 (Universal Harmonic Partition). Every stable relational system admits an orthogonal decomposition of its total frequency ω_N:

$$\omega_N^2 = \omega_V^2 + \omega_m^2, \tag{1}$$

where ω_V is the projection frequency (visible, relational mode), ω_m is the confinement frequency (inertial, informational mode), and $\omega_N = mc^2/\hbar$ for massive systems. This is treated as a physical constraint selecting stable configurations through rational ratios $\omega_V : \omega_m$. A derivation from more primitive axioms is an open question.

The notation $\bar{\lambda}_p = \hbar/(m_p c)$ denotes the reduced Compton wavelength of the proton, used throughout as the hadronic length scale. All frequency ratios below are derived from Postulate 1 and the binary closure defined in §2.2.

2.2 The Generative Hierarchy and Binary Closure

Before stating the closure postulate, we establish a motivation for why stable physical systems select specific closure depths. Consider the sequence of relational systems ordered by how many independent informational layers they encode (Table 3).

[POSTULATE] 2 (Binary Information Closure). A relational system achieves maximal stability when its harmonic partition satisfies a binary closure of depth n:

$$\frac{\omega_V^2}{\omega_N^2} = \frac{1}{2^n}, \qquad n \in \mathbb{N}, \tag{2}$$

ε	Depth n	$2^n - 1$ (factored)	Physical system	Status
0	0	—	Unrelated reference (photon, $\varepsilon = 0$)	No confinement
1	1	1	Related photon ($\varepsilon = 1$)	Trivial closure
2	2	3 (prime)	Electron family ($\varepsilon = 2$)	Single-prime closure
3	3	7 (prime)	Neutron ($\varepsilon = 3$)	Single-prime, unstable free
4	4	3×5	Proton ($\varepsilon = 4$)	First composite closure

Table 3: Generative hierarchy of closure depths. The column "$2^n - 1$ factored" shows the arithmetic structure of the confinement-to-visible ratio.

equivalently $\omega_V^2 : \omega_m^2 = 1 : (2^n - 1)$. The integer n counts the depth of binary subdivision in the information manifold.

Remark 2.1 — Structural motivation for $n = 4$. This manuscript does not claim a theorem fixing n. We record a two-part structural motivation.
(i) Arithmetic irreducibility. From Table 3, the proton is the first system whose confinement ratio $2^n - 1 = 15 = 3 \times 5$ is the product of two distinct primes. At $n = 2$ and $n = 3$, the ratio is prime (3 and 7 respectively), meaning the confinement mode has a single irreducible factor. At $n = 4$, both modes are arithmetically independent — neither can be expressed as a multiple of the other by an integer factor. This is one interpretation of why $n = 4$ represents the first genuinely stable closure: the two modes cannot collapse into a simpler shared resonance.
(ii) Representation-theoretic motivation. For a spin-$\frac{1}{2}$ system, the observable algebra is generated by Pauli matrices satisfying $(\hat{n} \cdot \boldsymbol{\sigma})^2 = I$, with $\hat{n}$ a unit vector. The fundamental representation of $\mathfrak{su}(2)$ is 2-dimensional over $\mathbb{C}$, hence 4-dimensional over $\mathbb{R}$. Under the reading that n encodes the minimal real degrees needed to store a coherent non-commutative spin description (reference state, measurement axis, non-commuting generator, binary orientation label), the baryonic assignment $n = 4$ is the first depth compatible with this real dimensionality.

Note on notation: $\hat{n}$ (unit vector) is distinct from the closure depth integer n throughout. A complete selection-rule theorem remains an open problem.

2.3 Identity Radius as Derived Quantity

[DERIVED] The visible radius is defined operationally as $R_V \equiv c/\omega_V$. For a nucleon in closure at depth $n = 4$, $\omega_V = \omega_N/2^{n/2} = \omega_p/4$, giving:

$$R_V^{(\mathrm{nuc})} = 4\bar{\lambda}_p = \frac{4\hbar}{m_p c} \approx 0.8412 \text{ fm}. \tag{3}$$

For a composite body of N nucleons: $R_{\mathrm{id}} = N^{1/3} R_V^{(\mathrm{nuc})}$.

[HYPOTHESIS] The numerical proximity to the CODATA 2018 proton charge radius (0.8414±0.0019 fm) and to the muonic hydrogen determination (0.84087±0.00039 fm) is treated as a falsifiable consistency check of the $n = 4$ assignment, not as a confirmed independent prediction.

3 Gravitational Model: Derivation from Harmonic Closure

3.1 Relational Distance as Phase Offset

Within this framework, one possible interpretation of the distance between two systems is as a relational phase offset: the accumulated difference in identity-mode frequency, measured in units of each system's identity radius. We state this as a hypothesis, acknowledging that other interpretations of the formalism may exist.

The identity radius R_{id} is the minimum path-length difference at which two systems of the same closure type first become distinguishable — not a geometric container, but a relational threshold. For two systems separated by coordinate distance r, one possible identification of the relational phase offset is:

$$\Phi(r) = \frac{r}{R_{\text{id}}}, \qquad R_{\text{id}} \equiv \frac{c}{\omega_V}. \tag{4}$$

$\Phi(r)$ is dimensionless: it counts how many identity-radii of difference separate the two systems. At $r = R_{\text{id}}$, the systems are one identity apart — the minimum distinguishable separation.

[POSTULATE] 3 (Relational Difference Elimination). Physical systems tend to reduce their mutual relational phase offset. One interpretation: the gravitational interaction is the macroscopic expression of this tendency — the large-scale dynamics of systems minimizing the informational distance between their projected states. This is one possible physical reading of the formalism; the mathematics does not uniquely select it.

How does a nucleon in the Sun produce a phase-offset field at planetary distances? In this interpretation, the identity mode of each nucleon projects outward via its visible cross-section R_V^2, establishing a relational field that any other nucleonic system intersects. The confinement depth R_m determines how deeply an incoming phase offset penetrates the receiving system's identity mode. Gravity, under this reading, is not mediated by a conventional field propagating through space — it is a feature of the relational structure itself: the accumulated phase-offset field of a massive body (a coherent sum over its nucleons) creates a background against which all other systems experience a tendency toward closure. The inverse-square law, derived in §3.2, quantifies how this tendency weakens with relational separation.

3.2 Conservation of Relational Information — Deriving the Inverse-Square Law

[POSTULATE] 4 (Isotropic Closure). The harmonic closure condition $\omega_N^2 = \omega_V^2 + \omega_m^2$ does not select a preferred spatial direction. Consequently, the relational phase offset of a source system propagates isotropically outward from its identity surface. A derivation of this isotropy from the generative algebraic structure of the framework remains future work.

[DERIVED] From Postulates 3 and 4, the total relational information emitted by a source is conserved over concentric surfaces of distinguishability:

$$\Delta\Phi(r) \cdot 4\pi r^2 = \Delta\Phi(R_{\text{id}}) \cdot 4\pi R_{\text{id}}^2 = \text{const}, \tag{5}$$

giving

$$\Delta\Phi(r) = \Delta\Phi(R_{\rm id}) \cdot \frac{R_{\rm id}^2}{r^2}. \tag{6}$$

This is the origin of the inverse-square law — derived from conservation of relational information over isotropic surfaces, not assumed.

3.3 The Gravitational Potential as Accumulated Phase Offset

[HYPOTHESIS] The gravitational potential $\phi(r)$ is identified, in this interpretation, as the energy cost of maintaining a relational phase offset at distance r:

$$\phi(r) = -\alpha\omega_m^2 \int_r^\infty \Delta\Phi(r')\, dr' \cdot \frac{M}{m_p}. \tag{7}$$

The factor ω_m^2 converts the dimensionless phase offset into an energy density; M/m_p counts the number of nucleonic sources. Here α is the same dimensionless coupling defined in §3.5.

[DERIVED] Substituting Eq. (6) and integrating: $\int_r^\infty R_{\rm id}^2/r'^2\, dr' = R_{\rm id}^2/r$, giving

$$\phi(r) = -\alpha\omega_m^2 \frac{R_{\rm id}^2}{m_p} \cdot \frac{M}{r}. \tag{8}$$

Dimensional check: $[\omega_m^2] \sim \mathrm{T}^{-2}$, $[R_{\rm id}^2] \sim \mathrm{L}^2$, $[m_p^{-1}] \sim \mathrm{M}^{-1}$; right-hand side $\sim \mathrm{L}^2\mathrm{T}^{-2}$. ✓

3.4 Relational Closure Volume and Source Density

To extend to a distributed source density ρ_m, we identify the relational closure volume of each nucleon — the volume through which it exchanges relational phase offset. The relational exchange is anisotropic between modes: the identity projects a transverse section R_V^2 toward other systems, while the confinement mode determines a penetration depth R_m in the orthogonal direction.

[HYPOTHESIS] Each nucleon in harmonic closure occupies a relational closure volume

$$v_{\rm id} = R_V^2 \cdot R_m, \tag{9}$$

where $R_V = c/\omega_V$ is the identity surface (the cross-section visible to another system) and $R_m = c/\omega_m$ is the confinement depth (how deeply the phase offset penetrates into the system's identity mode). This volume is not the geometric size of the nucleon — it is the effective volume through which relational phase offset is exchanged.

[DERIVED] The specific relational closure volume (per unit mass) is:

$$\kappa \equiv \frac{v_{\rm id}}{m_p} = \frac{R_V^2 R_m}{m_p} \approx 9.19 \times 10^{-20}\ \mathrm{m^3\,kg^{-1}}. \tag{10}$$

The relational source density and Poisson equation:

$$\rho_{\rm rel} \equiv \kappa\omega_m^2\rho_m, \tag{11}$$

$$\nabla^2\phi = 4\pi\alpha\rho_{\rm rel} = 4\pi\alpha\kappa\omega_m^2\rho_m. \tag{12}$$

Dimensional check: $[\kappa\omega_m^2\rho_m] = [\mathrm{L}^3\mathrm{M}^{-1}][\mathrm{T}^{-2}][\mathrm{ML}^{-3}] = [\mathrm{T}^{-2}]$. ✓

3.5 Newton's Constant as a Derived Quantity

[DERIVED] For a point mass M, Eq. (12) gives $\phi(r) = -\alpha\kappa\omega_m^2 M/r$. Comparing with Newton's potential $\phi_N(r) = -GM/r$:

$$G = \alpha\kappa\omega_m^2 = \alpha\frac{R_V^2 R_m}{m_p}\omega_m^2. \tag{13}$$

The confinement frequency ω_m and radius R_m are related by $R_m = c/\omega_m$, so $\omega_m^2 R_m = \omega_m c$. We define the centripetal acceleration at the confinement radius:

$$a_m \equiv \omega_m^2 R_m = \frac{c^2}{R_m} \approx 4.138 \times 10^{32}\ \mathrm{m/s^2}. \tag{14}$$

This is the centripetal acceleration intrinsic to the confinement mode — the rate at which the confinement boundary of a nucleon in $n = 4$ closure would need to accelerate to maintain its circular geometry. In the foundational documents, acceleration is defined as the mechanism that maintains the minimal identity coherence of each particle during interaction: $a = \omega^2 R$. The quantity a_m is precisely that mechanism at the confinement scale.

[DERIVED] Substituting $a_m = \omega_m^2 R_m$ into Eq. (13):

$$G = \alpha \cdot \frac{R_V^2}{m_p} \cdot a_m. \tag{15}$$

This separates G into three structurally independent factors, each with a physical interpretation within the framework:

1. α — the probability of relational phase transmission between two nucleons (double suppression through the visible mode);
2. R_V^2/m_p — the specific identity cross-section, the area that each nucleon's identity mode presents per unit mass;
3. a_m — the centripetal acceleration at the confinement radius, the intrinsic tendency of the confinement boundary to maintain coherence.

Numerical verification: $\alpha = 3.8124 \times 10^{-40}$, $R_V^2/m_p = 4.2309 \times 10^{-4}\ \mathrm{m^2/kg}$, $a_m = 4.1378 \times 10^{32}\ \mathrm{m/s^2}$. Product: $6.6743 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} = G$ (CODATA 2018). Error: 0.0000%.

Remark on the three centripetal accelerations of the $n = 4$ closure. The closure at depth $n = 4$ defines three natural centripetal accelerations:

$$a_N = \frac{c^2}{\bar{\lambda}_p} \approx 4.27 \times 10^{32}\ \mathrm{m/s^2} \quad \text{(Compton scale)},$$
$$a_V = \frac{c^2}{R_V} \approx 1.07 \times 10^{32}\ \mathrm{m/s^2} \quad \text{(identity scale)},$$
$$a_m = \frac{c^2}{R_m} \approx 4.14 \times 10^{32}\ \mathrm{m/s^2} \quad \text{(confinement scale)}.$$

Their ratios reproduce exactly the modal frequency ratios: $a_V/a_N = \omega_V/\omega_N = 1/4$, $a_V/a_m = \omega_V/\omega_m = 1/\sqrt{15}$. The geometric coupling $\alpha_{\text{geom}} = 1/(4\sqrt{15}) = (a_V/a_N)(a_V/a_m)$, the double suppression that produces the relational coupling, is therefore expressible as a product of two centripetal-acceleration ratios. This shows that α_{geom} — the geometric part of α — is completely determined by the internal acceleration structure of the $n = 4$ closure, not by any gravitational input.

3.6 Calibration and Parameter Accounting

For nucleonic matter in ground-state closure ($n = 4$), using CODATA 2018 values:

$$R_V = \frac{c}{\omega_V} = 4\bar{\lambda}_p \approx 8.412 \times 10^{-16}\ \text{m}, \tag{16}$$

$$R_m = \frac{c}{\omega_m} = \frac{4}{\sqrt{15}}\bar{\lambda}_p \approx 2.172 \times 10^{-16}\ \text{m}, \tag{17}$$

$$\kappa = \frac{R_V^2 R_m}{m_p} \approx 9.19 \times 10^{-20}\ \text{m}^3\,\text{kg}^{-1}, \tag{18}$$

$$\omega_m^2 = \frac{15}{16}\omega_p^2 \approx 1.905 \times 10^{48}\ \text{s}^{-2}. \tag{19}$$

[CALIBRATION] Setting $G = 6.67430 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ (CODATA 2018) fixes $\alpha \approx 3.81 \times 10^{-40}$. This is the single empirical input to the framework.

Geometric relation between the three radii. From the closure ratios at $n = 4$, one can show that R_V and R_m satisfy:

$$R_V^2 = 4\sqrt{15}\, R_N R_m, \tag{20}$$

where $R_N = \hbar/(m_p c) = \bar{\lambda}_p \approx 2.103 \times 10^{-16}$ m is the reduced Compton radius. This is a purely geometric consequence of the $n = 4$ partition: it holds identically and can be verified numerically ($R_V = 8.412 \times 10^{-16}$ m, $R_m = 2.172 \times 10^{-16}$ m, $R_N = 2.103 \times 10^{-16}$ m $\rightarrow R_V^2 = 7.08 \times 10^{-31}$ m^2). This relation eliminates α from the geometric structure and shows that the three length scales of the closure are not independent.

3.7 The Natural Gravitational Scale

[DERIVED] The product $\kappa\omega_m^2$ is constructible entirely from m_p, c, $\hbar$, and the $n = 4$ closure, with no gravitational input. Using $\bar{\lambda}_p^3\omega_p^2 = \hbar c/m_p$ (exact):

$$\kappa\omega_m^2 = \frac{R_V^2 R_m}{m_p}\omega_m^2 = 4\sqrt{15}\,\frac{\hbar c}{m_p^2} \approx 1.751 \times 10^{29}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{21}$$

This is the natural gravitational scale of the $n = 4$ closure:

$$\frac{G}{\alpha} = 4\sqrt{15}\,\frac{\hbar c}{m_p^2} = 2^{n/2}\sqrt{2^n - 1}\,\frac{\hbar c}{m_p^2}. \tag{22}$$

[DERIVED] Introducing the standard dimensionless gravitational coupling $\alpha_G \equiv Gm_p^2/(\hbar c)$, the coupling α admits an exact closed-form expression:

$$\alpha = \alpha_G \cdot \frac{\omega_V^2}{\omega_N \omega_m} = \frac{Gm_p^2}{\hbar c} \cdot \frac{1}{4\sqrt{15}} \approx 3.81 \times 10^{-40}. \tag{23}$$

Interpretation: α is the standard gravitational coupling α_G weighted by the relational ratio $\omega_V^2/(\omega_N \omega_m)$ — the structural proportion of the visible mode within the harmonic closure.

The smallness of α does not reflect an arbitrary hierarchy. It is the product of two suppressions: $\alpha_G \approx 5.9 \times 10^{-39}$ (the weakness of gravity at hadronic scales) and the relational ratio $\omega_V^2/(\omega_N \omega_m) = 1/(4\sqrt{15}) \approx 0.065$ — the fraction of the total closure structure that projects into the visible, relational sector. Within this interpretation, gravity is weak because the visible mode carries only $\omega_V^2/\omega_N^2 = 1/16$ of the total harmonic partition energy, and its coupling to the confinement mode is further attenuated by $\omega_V/\omega_m = 1/\sqrt{15}$.

Property	**Description**
Natural scale	$G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2 \approx 1.751 \times 10^{29}$ $\mathrm{m^3\,kg^{-1}\,s^{-2}}$. Fixed by $n = 4$ closure. No gravitational input.
α structural form	$\alpha = \alpha_G \cdot \omega_V^2/(\omega_N \omega_m)$. Relational ratio of the visible mode.
Geometric relation	$R_V^2 = 4\sqrt{15}\,R_N R_m$. Purely geometric; eliminates α from modal structure.
Empirical input	**[CALIBRATION]** One of $\{G, \alpha_G\}$ from CODATA 2018. Fixes $\alpha \approx 3.81 \times 10^{-40}$.
Open question	Deriving $\alpha_G = Gm_p^2/(\hbar c)$ from the closure would close the program.

Table 4: Parameter accounting (v4.0).

3.8 α_G as the Confinement Frequency Ratio

The standard gravitational coupling $\alpha_G \equiv Gm_p^2/(\hbar c) \approx 5.91 \times 10^{-39}$ is conventionally treated as an empirical constant. Within this framework, it admits a reinterpretation that places it inside the frequency hierarchy of the closure.

Step 1 — Gravitational instability of the torsional field. Define the gravitational instability parameter of the torsional field at scale R:

$$\eta(R) = \frac{E_{\text{grav}}(R)}{E_{\text{field}}} = \frac{G\hbar\omega^2}{c^5}, \tag{24}$$

where $E_{\text{field}} = \hbar\omega$ and $E_{\text{grav}}(R) = G(\hbar\omega)^2/(c^4 R)$. Evaluating at the nucleonic closure scale $R = \bar{\lambda}_p = c/\omega_N$:

$$\eta(\bar{\lambda}_p) = \frac{Gm_p^2}{\hbar c} = \alpha_G. \tag{25}$$

α_G measures how far the nucleon is from the Planck gravitational threshold. The torsional field is gravitationally stable ($\eta < 1$) for all $R > \bar{\lambda}_p$. Numerically: $\eta(\bar{\lambda}_p) = 5.9061 \times 10^{-39} = \alpha_G$. Ratio = 1.00000000

Step 2 — The Planck frequency from self-consistency. The condition $\eta = 1$, combined with $R = c/\omega$ (the field fills its coherence volume), yields:

$$R_{\text{crit}}^2 = \frac{G\hbar}{c^3} = \ell_{\text{Pl}}^2, \qquad \omega_{\text{Pl}} = \sqrt{\frac{c^5}{G\hbar}} \approx 1.855 \times 10^{43} \text{ rad/s}. \tag{26}$$

ω_{Pl} is the scale where the torsional field becomes gravitationally self-referential — where its self-energy equals its field energy. Consistent with the foundational framework's identification of ℓ_{Pl} as the minimum relational radius.

Step 3 — α_G as a squared frequency ratio. From steps 1–2:

$$\alpha_G = \left(\frac{\ell_{\text{Pl}}}{\bar{\lambda}_p}\right)^2 = \left(\frac{\omega_N}{\omega_{\text{Pl}}}\right)^2. \tag{27}$$

Expressing in terms of the confinement frequency ($\omega_N = (4/\sqrt{15})\omega_m$):

$$\alpha_G = \frac{16}{15}\left(\frac{\omega_m}{\omega_{\text{Pl}}}\right)^2. \tag{28}$$

The factor 16/15 is the inverse of the visible fraction of the $n = 4$ closure ($\omega_V^2/\omega_N^2 = 1/16$), a structural constant of the partition — not a free number. Numerically: $(1.6163 \times 10^{-35}/2.1031 \times 10^{-16})^2 = 5.906 \times 10^{-39}$

Remark 3.2 — Proposed logical order within the framework. Conventionally: G is empirical $\to$ $\alpha_G = Gm_p^2/(\hbar c)$ is computed. The framework supports inverting this: (1) ω_m is derived from the $n = 4$ closure; (2) ω_{Pl} emerges as the $\eta = 1$ self-consistency limit of the torsional field; (3) $\alpha_G = (\omega_N/\omega_{\text{Pl}})^2$ is their squared ratio; (4) $G = \alpha_G \cdot \hbar c/m_p^2$ follows. Under this reading, ω_{Pl} is more fundamental than G — it is the frequency where relational existence becomes self-referential — and α_G is not an arbitrary input but the measure of how far ω_N sits below ω_{Pl}. Status: this is a consistent reinterpretation within the framework, not a derivation. The open problem is to derive the ratio $\omega_{\text{Pl}}/\omega_N \approx 1.30 \times 10^{19}$ from the closure hierarchy without experimental input.

Result	Type	Numerical check
$\eta(\bar{\lambda}_p) = \alpha_G$ (Eq. 25)	[HYPOTHESIS]	$5.9061 \times 10^{-39} = 5.9061 \times 10^{-39}$
ℓ_{Pl} from $\eta = 1$ (Eq. 26)	[HYPOTHESIS]	1.6163×10^{-35} m (CODATA)
$\alpha_G = (\omega_N/\omega_{\text{Pl}})^2$ (Eq. 27)	[HYPOTHESIS]	$(1.616 \times 10^{-35}/2.103 \times 10^{-16})^2 = 5.906 \times 10^{-39}$
$\alpha_G = (16/15)(\omega_m/\omega_{\text{Pl}})^2$ (Eq. 28)	[HYPOTHESIS]	16/15 is structural constant of $n = 4$
Derive $\omega_{\text{Pl}}/\omega_N$ from closure	[OPEN]	$\varepsilon = 0 \to 4$ gives $O(1)$ ratios; insufficient

Table 5: Status of α_G reinterpretation (v4.0).

3.9 Summary: Four Structural Outputs from One Postulate

[DERIVED] The harmonic partition $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure ($n = 4$) yields four independent structural outputs from the same machinery:

1. **The inverse-square law** — derived from conservation of relational information over isotropic surfaces: $\Delta\Phi(r) \propto R_V^2/r^2 \Rightarrow |\mathbf{g}| \propto 1/r^2$.
2. **The natural gravitational scale** — purely from closure: $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$, fixed by m_p, c, $\hbar$, $n = 4$. No gravitational input.
3. G **reexpressed as** $G = \alpha \cdot R_V^2 \cdot a_m/m_p$ — three structurally independent factors (transmission probability, identity cross-section, confinement acceleration).
4. α_G **reinterpreted as** $(\omega_N/\omega_{\mathbf{Pl}})^2$ — the squared ratio of hadronic to Planck frequency, with ω_{Pl} emerging from the self-consistency condition of the torsional field.

4 Unification Across Scales

All four physical domains follow from $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure, spanning approximately 46 orders of magnitude (Table 6).

Scale	**System**	n	ω_V	ω_m	**Emergent property**
Hadronic	Proton	4	$\omega_p/4$	$\sqrt{15}/4\,\omega_p$	Charge radius ≈ 0.8412 fm
Atomic	s–p valence	2	N_s freq.	N_p freq.	Harmonic weight $P = a + b$
Cosmological	Hubble flow	0 (lim.)	H_0	$\to 0$	Expansion law $V = H_0 D$
Gravitational	Newtonian	4 (nuc.)	c/R_{id}	ω_m	$G = \alpha\kappa\omega_m^2$, $1/r^2$ derived

Table 6: Quadripartite unification via $\omega_N^2 = \omega_V^2 + \omega_m^2$. Spans ~ 46 orders of magnitude.

5 Testable Predictions and Falsifiability

5.1 Composition Dependence — Weak Equivalence Principle

[HYPOTHESIS] If ω_m (or $\kappa = R_V^2 R_m/m_p$) varies slightly with composition — e.g., via binding energy per nucleon — the model predicts small WEP violations. The Eötvös parameter is

$$\eta_{A,B} \approx \frac{\Delta\omega_m^2}{\omega_m^2} + \frac{\Delta\kappa}{\kappa} = \frac{\Delta\omega_m^2}{\omega_m^2} + 2\frac{\Delta R_V}{R_V} + \frac{\Delta R_m}{R_m}. \tag{29}$$

As an illustrative example, if ω_m^2 scales linearly with binding energy per nucleon B, i.e., $\Delta\omega_m^2/\omega_m^2 \approx \beta\,\Delta B/B$, then

$$\eta \approx \beta\,\frac{\Delta B}{B} + 2\frac{\Delta R_V}{R_V} + \frac{\Delta R_m}{R_m}. \tag{30}$$

A quantitative estimate of β requires a microphysical model from nuclear structure; this is left for future work.

FALSIFICATION CONDITION: This model predicts η should correlate with mean binding energy per nucleon, not only with atomic mass. Current limits from

MICROSCOPE [6] give $\eta \lesssim 10^{-15}$. The model is falsified if η shows no composition-binding-energy correlation at next-generation experiment precision (STE-QUEST, MICROSCOPE follow-on).

5.2 Scale-Dependent Gravity and Dark-Matter Phenomenology

[SPECULATIVE] In diffuse systems, if closure is partial, the effective $\kappa(r) = R_V^2(r)R_m(r)/m_p$ becomes scale-dependent, giving $G_{\text{eff}}(r) = \alpha\kappa(r)\omega_m^2$. **FALSIFICATION CONDITION:** Rotation-curve anomalies should correlate with informational density $\omega_m^2\rho_m$, not only with total mass distribution. This prediction distinguishes the framework from both ΛCDM and MOND [9, 10]. Important caveat: the r-dependence of κ is not derived from the postulates. This is a program note, not a genuine prediction.

6 Discussion

6.1 Comparison with Other Emergent Gravity Frameworks

Approach	**Source of gravity**	**Status of G**	**$1/r^2$ law**
Newton/Einstein	Mass / stress-energy	Fundamental	Assumed
Verlinde [4]	Entropy gradient	Proportionality constant	Assumed
Jacobson [3]	Horizon thermodynamics	Proportionality constant	Assumed
Padmanabhan [5]	Horizon/thermodynamics	Emergent (qual.)	Assumed
This work	Phase-offset density	$G = \alpha R_V^2 R_m \omega_m^2/m_p$	Derived

Table 7: Comparison with emergent-gravity frameworks. "Quantitative" refers to internal program consistency, not independent experimental confirmation.

Verlinde derives an entropic force $F = T\Delta S/\Delta x$ recovering Newton's second law near a holographic screen, where T is the Unruh temperature and ΔS is the entropy change. The approach is physically motivated, but T and ΔS are assigned to macroscopic holographic screens without a microscopic derivation of why G takes its observed value. Jacobson's derivation faces the same issue: the Einstein equation emerges from thermodynamics applied to local Rindler horizons, but G enters through the identification of entropy with horizon area, which is itself proportional to $1/G$. In contrast, the present framework assigns a specific relational closure volume κ to each nucleon, making G explicitly dependent on hadronic-scale physics. Critically, all prior approaches assume the $1/r^2$ form from the outset; this framework derives it.

6.2 The Mechanism: How a Nucleon Affects Distant Systems

A natural objection is: why does the internal structure of a proton in the Sun affect a trajectory 15 light-minutes away? The answer within this framework is not that the proton "reaches out" through a medium, but that distance itself is interpreted as a relational quantity. Each nucleon's identity mode projects outward via its visible cross-section R_V^2, establishing a relational phase-offset field. This field does not propagate in the conventional sense — it is a feature of the relational geometry between systems.

The inverse-square decay is then a consequence of the same isotropy that applies to any conserved quantity on a sphere: the total relational information is fixed, and it is

diluted geometrically over the area of concentric surfaces. The confinement depth R_m sets how deeply the incoming field penetrates a receiving system — it is the coupling coefficient between source and receiver.

This is analogous, in spirit, to quantum entanglement: correlations exist between spatially separated systems without a propagating signal. The relational phase offset is a feature of the joint configuration of the two systems, not a mediated interaction. The inverse-square decay reflects the geometric dilution of relational information over isotropic surfaces, not the propagation of a force through a medium. We emphasize that this is one possible physical interpretation of the mathematics. The formalism is compatible with other readings, including purely geometric ones that do not require an explicit medium. What the framework does establish unambiguously is the functional form: inverse-square law with Newton's constant expressible in hadronic terms.

6.3 On the Notation of α

Throughout this paper, α denotes the universal dimensionless coupling defined in §3.5: $\alpha \approx 3.81 \times 10^{-40}$. This is distinct from $\alpha_G = Gm_p^2/(\hbar c) \approx 5.91 \times 10^{-39}$ (the standard gravitational coupling) and from the fine-structure constant $\alpha_{\text{em}} \approx 1/137$ (electromagnetic). The relation between them is exact: $\alpha = \alpha_G/(4\sqrt{15})$. All uses of α in Eqs. (7), (8), (12), (13) refer to the same calibrated quantity.

6.4 Limitations and Future Work

Limitations.

- Weak-field, stationary regime only.
- α is calibrated, not derived.
- Relativistic completion remains future work.
- Isotropy of Postulate 4 is stated, not derived from the algebra.
- Physical correspondence with QFT not established.
- n-selection is a structural motivation, not a proof.

Future work.

- Derive α_G from relational first principles. Version 4.0 establishes that $\alpha_G = \eta(\bar{\lambda}_p)$ (gravitational instability of the torsional field at hadronic scale) and $\alpha_G = (\omega_N/\omega_{\text{Pl}})^2$ (squared ratio of hadronic to Planck frequency). The open problem is to derive the value of $\omega_{\text{Pl}}/\omega_N \approx 1.30 \times 10^{19}$ from the closure hierarchy without experimental input. The levels $\varepsilon = 0 \to 4$ produce $O(1)$ ratios; a cosmological or collective-closure argument may be required.
- Derive isotropy of Postulate 4 from the generative algebra.
- Develop a strong-field tensor formulation.
- Quantitatively fit galaxy rotation curves using $\kappa(r)$.

- Establish the n-selection theorem.
- Derive finite-size corrections to the phase-offset integral for $r \lesssim R_{\mathrm{id}}$.
- Map nuclear binding energy to ω_m and R_m variation for a quantitative WEP prediction.

7 Conclusion

We have presented a derivation of Newtonian gravity from relational projection principles in which neither the inverse-square law nor Newton's constant are assumed. The key elements are:

1. **[DERIVED]** Relational phase offset $\Phi(r) = r/R_{\mathrm{id}}$ as the fundamental measure of separation (Postulate 3).
2. **[DERIVED]** Conservation of relational information over isotropic surfaces $\Rightarrow$ $\Delta\Phi(r) \propto R_{\mathrm{id}}^2/r^2 \Rightarrow$ inverse-square law (Postulates 3–4). Derived, not assumed.
3. **[DERIVED]** Gravitational potential as accumulated phase offset $\Rightarrow \phi(r) \propto -M/r$.
4. **[DERIVED]** Relational closure volume $\kappa = R_V^2 R_m/m_p$ from the two modal scales of the harmonic partition.
5. **[DERIVED]** Poisson equation $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$ with $G = \alpha\kappa\omega_m^2$.
6. **[DERIVED]** Newton's constant reexpressed: $G = \alpha \cdot R_V^2 \cdot a_m/m_p$, where $a_m = c^2/R_m$ is the centripetal acceleration at the confinement radius. Three factors: relational transmission probability, specific identity cross-section, confinement acceleration.
7. **[DERIVED]** Natural gravitational scale: $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$, fixed entirely by m_p, c, $\hbar$, $n = 4$. No gravitational input required.
8. **[DERIVED]** Geometric relation $R_V^2 = 4\sqrt{15}\,R_N R_m$. The acceleration ratios $(a_V/a_N)(a_V/a_m) = 1/(4\sqrt{15})$ reproduce the geometric coupling exactly from internal accelerations.
9. **[DERIVED]** Structural form of α: $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N\omega_m)$ — the relational ratio of the visible mode within the $n = 4$ closure.
10. **[HYPOTHESIS]** $\alpha_G = \eta(\bar{\lambda}_p)$ — gravitational instability of the torsional field at hadronic scale. $\alpha_G = (\omega_N/\omega_{\mathrm{Pl}})^2 = (16/15)(\omega_m/\omega_{\mathrm{Pl}})^2$. The Planck frequency emerges as the $\eta = 1$ self-consistency threshold. Numerically verified: $\eta(\bar{\lambda}_p) = \alpha_G = 5.906 \times 10^{-39}$ (numerical, CODATA 2018).
11. **[MOTIVATED]** $n = 4$ as the first closure depth where $2^n - 1 = 3 \times 5$ is arithmetically composite (two distinct prime factors), suggesting irreducible incommensurability. A complete selection-rule theorem remains open.
12. **[HYPOTHESIS]** Falsifiable avenues: composition-dependent WEP deviations correlating with binding energy per nucleon; scale-dependent effective G distinguishable from MOND in rotation curves.

13. **[CALIBRATION]** One empirical input: $\alpha_G = Gm_p^2/(\hbar c)$ from CODATA 2018. Version 4.0 reinterprets this as $(\omega_N/\omega_{\mathrm{Pl}})^2$ — the program's primary open question is now to derive this ratio from closure.

    Gravity, like the size of the proton, the stability of atoms, and the expansion of the cosmos, is not a fundamental interaction but a manifestation of relational geometry seeking harmonic closure — the macroscopic tendency of systems to reduce the phase offset that separates them from each other.

Notation and Glossary

Symbol	Value / Form	Meaning
ω_N	$m_p c^2/\hbar$	Total (Compton) frequency of the system.
ω_V	$\omega_N/2^{n/2}$	Visible mode: the relational, projective frequency.
ω_m	$\omega_N\sqrt{1-2^{-n}}$	Confinement mode: the inertial, identity frequency.
$\bar{\lambda}_p$	$\hbar/(m_p c)$	Reduced Compton wavelength of the proton ≈ 0.2103 fm.
R_V	$c/\omega_V = 4\bar{\lambda}_p$	Identity radius (transverse cross-section) ≈ 0.8412 fm.
R_m	$c/\omega_m = 4\bar{\lambda}_p/\sqrt{15}$	Confinement depth (longitudinal penetration) ≈ 0.2172 fm.
R_N	$\hbar/(m_p c) = \bar{\lambda}_p$	Reduced Compton radius ≈ 0.2103 fm.
κ	$R_V^2 R_m/m_p$	Specific relational closure volume $\approx 9.19 \times 10^{-20}$ m^3 kg^{-1}.
α	$\alpha_G \cdot \omega_V^2/(\omega_N\omega_m)$	Universal dimensionless coupling $\approx 3.81 \times 10^{-40}$.
α_G	$Gm_p^2/(\hbar c)$	Standard dimensionless gravitational coupling $\approx 5.91 \times 10^{-39}$. Version
a_m	c^2/R_m $(= \omega_m^2 R_m)$	Centripetal acceleration at the confinement radius $\approx 4.138 \times 10^{32}$ m/s
$\Phi(r)$	r/R_{id}	Relational phase offset at separation r. Dimensionless.
n	4 (baryonic)	Binary closure depth. Fixes $\omega_V/\omega_N = 1/2^{n/2}$, $\omega_m/\omega_N = \sqrt{1-2^{-n}}$.
$\hat{n}$	unit vector	Unit vector in Pauli identity $(\hat{n}\cdot\boldsymbol{\sigma})^2 = I$. Distinct from closure depth

Table 8: Notation glossary. All quantities for nucleonic matter in ground-state closure ($n = 4$), CODATA 2018.

LLM Disclosure Statement

Models used: Anthropic Claude, OpenAI ChatGPT, Google Gemini, and DeepSeek AI.

How they were employed:

- **DeepSeek AI**: adversarial cross-validation (AACV) of algebraic derivations, stress-testing the n-selection argument, and verifying dimensional consistency across all equations.
- Dimensional-analysis consistency checking (all models).
- Algebraic stress-testing: presenting each postulate chain and requesting identification of unjustified steps or circular reasoning.
- Literature synthesis: identifying prior work in emergent gravity and equivalence-principle experiments.
- Adversarial Cross-Validation (AACV): separate sessions prompted with neutral review tasks.
- Structural refinement in response to community critique (r/LLMPhysics): the derivation of $1/r^2$ from conservation of relational information, the redefinition of κ as $R_V^2 R_m/m_p$, the n-selection hierarchy (Table 3), and the clarification of the gravitational mechanism were all developed in response to reviewer feedback.
- Language revision and LaTeX formatting.

Estimated contribution: $\sim 30\%$ drafting/structure; $\sim 70\%$ human-directed conceptual development, postulate formulation, and iterative refinement.

Human oversight: All mathematical claims were verified by the author independently. LLM outputs were treated as drafting aids and consistency probes, not epistemic authorities. Citation details were cross-checked against primary sources.

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